# Title : Reducing Hallucination in Financial AI Through Retrieval - Augmented Agentic

*7-Week Capstone Project Management Plan*

Methodology: Scrum / Kanban | Jira Issue Tracking | GitHub Code Review

## Introduction

Project B.A.N.A.N.A (Brokerage Analytic Network & Autonomous Node Assistant), is a 7-week academic capstone project focused on building a production-grade Retrieval-Augmented Generation (RAG) system for grounded financial query answering. The system is designed to answer natural language questions about stocks, SEC filings, and market sentiment — while actively preventing hallucination, the tendency of large language models to fabricate plausible-sounding but factually incorrect information.

The core challenge this project addresses is trust in AI-generated financial information. A model that confidently states a wrong stock price or a fabricated earnings figure can cause real harm. Banana solves this by grounding every answer in verified source documents retrieved at query time from three live data providers — Alpha Vantage for market prices, SEC EDGAR for regulatory filings, and Twitter/X for social sentiment — and blocking any response that cannot be traced back to a retrieved source.

The system is built on a three-layer architecture. The outer layer ingests data through three Model Context Protocol (MCP) servers. The middle layer runs an 11-node LangGraph orchestration pipeline with three specialised agents — AnalystAgent (FinBERT for sentiment), ReflectionAgent (confidence gating), and ScribeAgent (report generation). The inner layer stores and retrieves documents using a ChromaDB vector store backed by a sentence-transformer embedding model.

The three-member team — **Gangadhar Shiva (Orchestration Lead Engineer), Ananya Chandraker (MCP Lead Engineer), and Akshobhya Rao BV (Agentic AI Lead Engineer) — manages the project using a Scrum/Kanban hybrid methodology and Gantt Chart. Work is tracked in Jira across seven one-week sprints, all code changes go through GitHub Pull Requests with mandatory peer review, and no code merges to the main branch without passing the hallucination probe test suite**. This structured approach ensures that a complex, multi-component ML system can be built, tested, deployed on Kubernetes, and fully documented within a compressed seven-week timeline.

## 1. Gantt Chart — 7-Week Timeline

Each row shows one deliverable, its owner, and the weeks it spans. Blue = pipeline/code (Ganga), Green = report/writing (Ananya), Amber = data/QA (Akshobhya), Purple = shared reviews.

| Task | Owner | W1 | W2 | W3 | W4 | W5 | W6 | W7 |
|---|---|---|---|---|---|---|---|---|
| Repo + Jira board setup | Ganga | | | | | | | |
| MCP servers (market, SEC, social) | Akshobhya | | | | | | | |
| FastAPI skeleton + /analyze endpoint | Ganga | | | | | | | |
| LangGraph pipeline (11 nodes) | Ganga | | | | | | | |
| Kubernetes deploy + Docker images | Ganga | | | | | | | |
| Agent integration (FinBERT/Scribe) | Ganga | | | | | | | |
| Hallucination fixes + ticker_guard | Ganga | | | | | | | |
| Test suite + hallucination probes | Akshobhya | | | | | | | |
| Benchmark runs + fix | Akshobhya | | | | | | | |
| Twitter/X social server (VADER) | Akshobhya | | | | | | | |
| Report: Intro + Architecture | Ananya | | | | | | | |
| Report: EDA + Pipeline Design | Ananya | | | | | | | |
| Report: Hallucination + Deployment | Ananya | | | | | | | |
| Report: Benchmark + Conclusion | Ananya | | | | | | | |
| Code reviews (PR #1–#6) | Ganga | | | | | | | |
| Writing reviews (sections 1–all) | Akshobhya | | | | | | | |
| Presentation slides | Ananya | | | | | | | |
| Record presentation video | Ganga | | | | | | | |
| Final report assembly | Ananya | | | | | | | |
| Final submission | Ganga | | | | | | | |

**Ganga — Pipeline/Code/Integration** | **Ananya — Report/Writing/Integration** | **Akshobhya — Data/QA/Design/Integration** | **Shared Reviews**

## 2. Jira Sprint Plan (BAN-001 – BAN-046)

> *Review rotation: Ganga reviews Akshobhya's code → Akshobhya reviews Ananya's writing → Ananya reviews Ganga's code. Self-review also permitted.*

### Sprint 1 · Goal: Foundation + Data Sources

| ID | Task | Owner | Reviewer | Dates | Status |
|---|---|---|---|---|---|
| BAN-001 | GitHub repo, branch strategy, .gitignore | Ganga | Akshobhya | Mar 6–6 | DONE |
| BAN-002 | Jira board, 7 sprints, epic labels | Ganga | Ananya | Mar 6–7 | DONE |
| BAN-003 | MCP market server (Alpha Vantage) | Akshobhya | Ganga | Mar 7–8 | DONE |
| BAN-004 | MCP SEC EDGAR server | Akshobhya | Ganga | Mar 8–9 | DONE |
| BAN-005 | MCP social server (placeholder stub) | Akshobhya | Ganga | Mar 9–9 | DONE |
| BAN-006 | FastAPI skeleton + /analyze endpoint | Ananya | Akshobhya | Mar 9–10 | DONE |
| BAN-007 | Report: Abstract + Introduction (draft) | Ananya | Akshobhya | Mar 9–11 | TO DO |
| BAN-008 | Code review PR #1 — MCP servers | Ganga | Akshobhya | Mar 11–12 | TO DO |

### Sprint 2 · Goal: LangGraph Pipeline

| ID | Task | Owner | Reviewer | Dates | Status |
|---|---|---|---|---|---|
| BAN-009 | LangGraph nodes: intent, fetch, validate, store, retrieve | Ganga | Akshobhya | Mar 13–15 | TO DO |
| BAN-010 | evaluate_node + answer_node + graph compile | Ganga | Akshobhya | Mar 15–15 | TO DO |
| BAN-011 | Kubernetes manifests + Docker images (all 4) | Ganga | Akshobhya | Mar 15–16 | TO DO |
| BAN-012 | test_service.py — 14 base queries, 5 categories | Akshobhya | Ananya | Mar 13–16 | TO DO |
| BAN-013 | Report: System Architecture section | Ananya | Akshobhya | Mar 16–17 | TO DO |
| BAN-014 | Report: Dataset & EDA section (draft) | Ananya | Akshobhya | Mar 17–18 | TO DO |
| BAN-015 | Code review PR #2 — LangGraph + K8s | Akshobhya | Ganga | Mar 18–19 | TO DO |

## Sprint 3 ·· Goal: Agents + Hallucination Fixes

| ID | Task | Owner | Reviewer | Dates | Status |
|---|---|---|---|---|---|
| BAN-016 | AnalystAgent (FinBERT) integration | Ganga | Ananya | Mar 20–21 | TO DO |
| BAN-017 | ReflectionAgent + confidence gate (0.70) | Ganga | Ananya | Mar 21–21 | TO DO |
| BAN-018 | ScribeAgent + conditional routing | Ganga | Akshobhya | Mar 21–22 | TO DO |
| BAN-019 | Add 3 hallucination probe queries to test suite | Akshobhya | Ananya | Mar 22–23 | TO DO |
| BAN-020 | Fix: social sentinel bypass (not_placeholder) | Ganga | Akshobhya | Mar 23–23 | TO DO |
| BAN-021 | Fix: ticker_guard_node (cross-ticker contamination) | Ganga | Akshobhya | Mar 23–24 | TO DO |
| BAN-022 | Report: Pipeline Design + Agent sections | Ananya | Akshobhya | Mar 24–25 | TO DO |
| BAN-023 | Code review PR #3 — agents + fixes | Ananya | Ganga | Mar 25–26 | TO DO |

## Sprint 4 · Goal: Deployment + Benchmark

| ID | Task | Owner | Reviewer | Dates | Status |
|---|---|---|---|---|---|
| BAN-024 | Minikube deploy, port-forward, smoke test | Ganga | Akshobhya | Mar 27–28 | TO DO |
| BAN-025 | Run test_service.py (17 queries incl. probes) | Akshobhya | Ananya | Mar 28–29 | TO DO |
| BAN-026 | Fix benchmark.py empty-doc bug + run 2 passes | Akshobhya | Ganga | Mar 29–30 | TO DO |
| BAN-027 | Report: Hallucination Prevention section | Ananya | Akshobhya | Mar 30–31 | TO DO |
| BAN-028 | Report: Deployment section | Ananya | Akshobhya | Mar 31–Apr 1 | TO DO |
| BAN-029 | Code review PR #4 — deployment + benchmark | Akshobhya | Ganga | Apr 1–2 | TO DO |

## Sprint 5 · Goal: Twitter/X Social Server + Report

| ID | Task | Owner | Reviewer | Dates | Status |
|---|---|---|---|---|---|
| BAN-030 | Twitter/X API v2 integration (social_server.py) | Akshobhya | Ganga | Apr 3–4 | TO DO |
| BAN-031 | VADER sentiment scoring + Dockerfile.social update | Akshobhya | Ganga | Apr 4–5 | TO DO |

| ID | Task | Owner | Reviewer | Dates | Status |
|---|---|---|---|---|---|
| BAN-032 | K8s secret + banana-social-deployment.yaml | Akshobhya | Ganga | Apr 5–5 | TO DO |
| BAN-033 | Code review PR #5 — social server | Ganga | Akshobhya | Apr 5–6 | TO DO |
| BAN-034 | Report: Benchmark Analysis section | Ananya | Akshobhya | Apr 6–7 | TO DO |
| BAN-035 | Report: Testing & Evaluation section | Ananya | Ganga | Apr 7–8 | TO DO |
| BAN-036 | Writing review — all report sections | Akshobhya | Ananya | Apr 8–9 | TO DO |

## Sprint 6 · Goal: Final Report + Presentation

| ID | Task | Owner | Reviewer | Dates | Status |
|---|---|---|---|---|---|
| BAN-037 | Report: Conclusion + Future Work + Appendix | Ananya | Akshobhya | Apr 10–11 | TO DO |
| BAN-038 | Final report assembly + docx formatting | Ananya | Ganga | Apr 11–12 | TO DO |
| BAN-039 | Full team review — code + final report | Ganga | Ananya | Apr 12–13 | TO DO |
| BAN-040 | Presentation slides (12 slides) | Ananya | Akshobhya | Apr 10–11 | TO DO |
| BAN-041 | Record presentation video (~10 min) | Ganga | Ananya | Apr 11–12 | TO DO |
| BAN-042 | Address review feedback, final revisions | Ganga | Akshobhya | Apr 12–13 | TO DO |
| BAN-043 | Code review PR #6 — final integration | Akshobhya | Ganga | Apr 10–11 | TO DO |

## Sprint 7 · Buffer + Submission

| ID | Task | Owner | Reviewer | Dates | Status |
|---|---|---|---|---|---|
| BAN-044 | Re-record presentation if needed | Ganga | Ananya | Apr 10–11 | TO DO |
| BAN-045 | Final proofreading — all 3 members | Ananya | Ganga | Apr 11–12 | TO DO |
| BAN-046 | Upload deliverables + final submission | Ganga | Akshobhya | Apr 12–13 | TO DO |

## 3. Jira Kanban Board — Sprint 1 Snapshot

*Project B.A.N.A.N.A · Mar 6–12, 2026 · Goal: Foundation + Data Sources + Report Start*

> *Sprint Goal: Set up the project foundation — GitHub repo, Jira board, all 3 MCP servers, FastAPI skeleton, and first report draft. All 8 issues target DONE by Mar 12.*

### 📋 BACKLOG — 0 issues

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### 📌 TO DO — 0 issues

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### ⚙️ IN PROGRESS — 0 issues

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### 👁️ IN REVIEW — 1 issue

**REVIEW**
**BAN-008**
Code review PR #1 — MCP servers
📅 Mar 11
Owner: **Ganga** | Reviewer: **Akshobhya**

### ✅ DONE — 7 issues

**CODE**
**BAN-001**
GitHub repo + branch strategy
📅 Mar 6
Owner: **Ganga** | Reviewer: **Akshobhya**

**CODE**
**BAN-002**
Jira board — 7 sprints + epics
📅 Mar 6–7
Owner: **Ganga** | Reviewer: **Ananya**

**CODE**
**BAN-003**
MCP market server (Alpha Vantage)
📅 Mar 7–8
Owner: **Akshobhya** | Reviewer: **Ganga**

**CODE**
**BAN-004**
MCP SEC EDGAR server
📅 Mar 8–9
Owner: **Akshobhya** | Reviewer: **Ganga**

**CODE**
**BAN-005**
MCP social server (stub)

### Legend

**Card Types**
- CODE
- REPORT
- REVIEW

**Members**
- Ganga
- Ananya
- Akshobhya

**Sprint Stats**
Total issues: **8**
Done: **7**
In Review: **1**
Velocity: **8 pts**

📅 Mar 8

| Owner | Reviewer |
|---|---|
| Akshobhya | Ganga |

CODE
BAN-006
FastAPI skeleton + /analyze
📅 Mar 9

| Owner | Reviewer |
|---|---|
| Ganga | Akshobhya |

REPORT
BAN-007
Report: Abstract + Introduction
📅 Mar 9

| Owner | Reviewer |
|---|---|
| Ananya | Akshobhya |

> *Sprint 1 end state: 7 of 8 issues DONE. BAN-008 (PR #1 code review) moves to DONE once Akshobhya completes the review on Mar 12. All deliverables on track — no rollover issues into Sprint 2.*

## 4. GitHub Code Review + Writing Review Workflow

| PR | Sprint | Scope | Author | Reviewer | Date |
|---|---|---|---|---|---|
| PR #1 | Sprint 1 | MCP servers (market, SEC, social) | Akshobhya | Ganga | Mar 11 |
| PR #2 | Sprint 2 | LangGraph pipeline + K8s manifests | Ganga | Akshobhya | Mar 18 |
| PR #3 | Sprint 3 | Agent integration + hallucination fixes | Ganga | Ananya | Mar 25 |
| PR #4 | Sprint 4 | Deployment + benchmark fix | Akshobhya | Ganga | Apr 1 |
| PR #5 | Sprint 5 | Twitter/X social server | Akshobhya | Ganga | Apr 5 |
| PR #6 | Sprint 6 | Final integration + bug fixes | Ganga | Akshobhya | Apr 13 |
| Writing R1 | Sprint 3–4 | Report sections 1–5 | Ananya | Akshobhya | Apr 2 |
| Writing R2 | Sprint 5–6 | Report sections 6–end + final | Ananya | Akshobhya | Apr 9 |

## 5. Presentation Recording Plan

| # | Slide Content | Presenter | Key Point | Time | Slide Due |
|---|---|---|---|---|---|
| 1 | Title + team + project goal | Ganga | Introduce B.A.N.A.N.A and the problem | 30s | Apr 13 |
| 2 | Problem — hallucination in finance | Ananya | Why fabricated data = real risk | 45s | Apr 13 |
| 3 | Architecture diagram (3-layer) | Ganga | Peel → Pulp → Core | 1m | Apr 13 |
| 4 | LangGraph pipeline — 11 nodes | Ganga | Each node and its guard function | 1m30s | Apr 13 |
| 5 | Agent integration (FinBERT/Scribe) | Ananya | How agents chain after answer_node | 1m | Apr 13 |
| 6 | Hallucination prevention — 7 layers | Ananya | Defence-in-depth | 1m | Apr 13 |
| 7 | 2 vulnerabilities found and fixed | Akshobhya | Probe results before/after | 1m | Apr 13 |
| 8 | Benchmark results Run 1 vs Run 2 | Akshobhya | Bug, fix, and 1.38x speedup | 1m30s | Apr 13 |
| 9 | Live demo — curl /analyze AAPL | Ganga | Full API response with sentiment | 1m | Apr 13 |
| 10 | Kubernetes deployment | Akshobhya | kubectl get pods output | 30s | Apr 13 |
| 11 | Limitations + future work | Ananya | Stale docs, VADER mismatch | 30s | Apr 13 |
| 12 | Conclusion | Ganga | 0% hallucination, 1.38x faster | 30s | Apr 13 |

> *Recording schedule: Ganga records video Apr 13 with Ananya and Akshobhya on a review call. Buffer re-record: Apr 13 (BAN-044). Final upload: Apr 13.*

## 5. Conclusion

This document represents the current project plan for B.A.N.A.N.A, outlining the structured approach the team will follow to successfully design, implement, and test the system over the seven-week capstone timeline. Rather than describing completed work, this plan serves as a roadmap that defines the technical architecture, sprint milestones, agent responsibilities, integration checkpoints, and validation mechanisms that will guide development.

By adhering to this structured plan — including weekly sprint goals, Jira-based task tracking, peer-reviewed GitHub pull requests, hallucination probe testing before merges, and incremental Kubernetes deployment — the team will ensure steady progress, controlled risk, and measurable quality at each stage. The clear division of responsibilities among engineering, machine learning, and data roles further supports accountability and efficient collaboration.

Following this plan will enable the team to systematically build a production-grade, hallucination-resistant financial RAG system within the compressed seven-week schedule. With disciplined execution, continuous testing, and structured evaluation metrics, B.A.N.A.N.A will progress from architectural design to a fully implemented and validated system capable of delivering grounded and trustworthy financial insights.